

Nimish Sharma

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About me

I am a fifth-year undergraduate student at the [Department of Physical Sciences](#) at the [Indian Institute of Science Education and Research, Kolkata](#), with a strong passion for quantum optics and photonics. Along with my academic pursuits, I have developed significant expertise in computational physics, and I wish to expand my domain into quantum and nonlinear optics. I thrive in collaborative environments and enjoy working in teams to solve complex scientific problems. Beyond academics, I am an avid football player and guitarist, always seeking a balance between analytical thinking and creative expression.

Research Interest

Photonics

- I am interested in exploring how light interacts with different materials, especially through the fields of photonics and metasurfaces. My recent work involved analysis of the Mueller matrix of a waveguide plasmonic crystal using a dark-field polarization microscopy setup to infer spin-orbit interactions of light and its implications, such as spin-orbit-spin coupling and spin hall effect. This gave me a solid foundation in experimental optics and polarization analysis, which I hope to build on further.
- I enjoy using optical techniques to uncover hidden material properties, and I want to bridge the gap between simulation and experiment. I use computational tools like MATLAB and Python to create models that theoretically explain experimental results and also help guide the experiments themselves. I believe combining simulations with hands-on lab work leads to a deeper and more accurate understanding.

Quantum and Non-Linear Optics

- I also want to move into the field of quantum optics and nonlinear optics. The vastness of research areas in quantum optics excites me, like chiral quantum optics, including on-chip quantum communication. I am interested in exploring quantum sensing using various techniques such as NV centres and optically pumped magnetometers.
- One area that excites me a lot is engineering quantum light sources using single-photon and multi-photon emitters based on quantum dots. These methods, along with fluorescence microscopy, offer powerful ways to look inside living systems at extremely small scales. I see huge potential for applying advanced optical and quantum tools to biomedical imaging, helping us understand biological processes better and possibly improve diagnostics.
- I have also been exploring fibre-based nonlinear optics through some courses. I am curious to explore the domain of fibre optics in the context of frequency conversion, four-wave mixing and optical parametric amplification. I have also been reading up on dual-comb spectroscopy and tailoring dispersion profiles via photonic crystals to enable phase matching. I see great potential in the recent research on exploiting the $\chi^{(2)}$ and $\chi^{(3)}$ nonlinear mediums for integrated photonics.
- Overall, I want to bring together computational physics, experimental optics, and quantum technologies to push the boundaries of how we study light and matter. I'm especially motivated by the real-world impact this work can have, whether it's in building better photonic devices or advancing imaging techniques.

Relevant Courses

- | | |
|---|--|
| • Advanced Electricity, Magnetism, and Optics | • Quantum Mechanics - I, II, III |
| • Advanced Optics Laboratory | • Linear Algebra - I |
| • Chemistry of Materials | • Programming and Data Structures in C |
| • Quantum Information Processing | • Computational Physics |
| • Biophysics II | • Data Structures and Algorithms in C |
| • Condensed Matter Physics | • Spectrochemistry & Organic Synthesis Lab |
| • Field Theory and Relativistic Quantum Mechanics | • Nuclear Physics Lab |
| • Condensed Matter Laboratory | • Optics and Modern Physics Lab |
| • Numerical Analysis Methods - MATLAB | • Electronics Lab |

Education

Indian Institute of Science Education and Research Kolkata

August 2021 – May 2026

- BS-MS in Physics along with a minor in Computer Science.
- CGPA(Current) : 8.1

Experience

Bio-NaP Lab | IISER Kolkata

January 2025 - Present

- Joined [Bio-NaP lab](#) at IISER Kolkata and currently working on dark field polarization microscopy setup.
- Integrating MATLAB simulations for the far-field Mueller matrix measurements and confirming experimental results, inferring spin-orbit interaction effects.
- Involved in experimental research on light-matter interaction of various metasurfaces and analysis of spectral and momentum domain Mueller matrices for the same.
- Investigating applications of plasmonic interactions for enhanced imaging and sensing technologies.

First Principle Investigations on Heusler Alloys | IAS SRFP – SVNIT Surat

May 2024 – July 2024

- Basics of Density Functional Theory and solid state physics were studied to grasp the theoretical understanding. It included reading research papers on Heusler alloys to understand their properties and structure.
- To get familiarised and gain confidence with the DFT software ([QE](#) and [WIEN2k](#)), results were replicated from a [paper on Ti2VGe](#) and the half-metallic nature was confirmed.
- Proceeding from the success, further first-principle calculations were performed for our chosen alloy. Structural optimisations, magnetic properties, elastic properties, dynamical stability and electronic properties were calculated.
- Finally, I compiled and submitted a report detailing DFT theory, software usage, and findings to the Indian Academy of Sciences.

Quantum Computing | Summer Internship – IISER Bhopal

May 2023 – July 2023

- Quantum Computing basics, including introduction to quantum gates, entanglement, quantum teleportation, and algorithms.
- Building a binary classifier for the Iris dataset by developing a Quantum Deep Learning Network.
- Tested the network on IBMQManila and compared its results with various Classical Machine Learning models such as Logistic Regression, KNeighbour and Decision Tree classifier.
- Creating a presentation and writing a report to conclude the results and learnings.

Integration of Payment Portal | Summer Project – IISER Kolkata

May 2023 – July 2023

- Built an Integration for a Payment Gateway into the existing mess website for my Institute.
- Used Django framework and SQLite. Learned front-end webpage designing using CSS and Javascript.
- Finally, I wrote a report concluding the work and things learned during the project.

Workshops

Nonlinear Fiber Optics | IKA

July 2025 - present

- The course lays a solid foundation on the theory and simulations of how light behaves inside optical fibers, starting right from Maxwell's equations.
- I learned about various nonlinear effects like self-phase modulation and four-wave mixing, and how they influence pulse propagation in fibers.
- We used tools like MATLAB and MEEP to simulate light behaviour in fibers, which helped connect theory with hands-on application.
- Overall, the course built a solid foundation in nonlinear fiber optics, with interesting real-world applications like communication systems and radar signal distribution.

Physics through Computational Thinking | NPTEL – IISER Bhopal

Feb 2023 - April 2023

- Wolfram Alpha and Fortran programming language.

- Introduction to Monte-Carlo Simulations and simulating statistical models.
- Application of Numerical Methods (Euler, Runge Kutta(2,4), etc.)

From Atoms to Materials: Predictive Theory and Simulations | [edX – Purdue University](#)

December 2023

- Relating the application of Classical Mechanics, Quantum Mechanics and Statistical Mechanics to the Real World.
- Online DFT and running MD simulations(on NanoHub) on basic molecules and predicting their thermal and electronic properties.

Lunar Exploration through Artificial Intelligence | Spartificial

May 2022 - June 2022

- Introduction to various Unsupervised Machine Learning Algorithms.
- It also familiarised me with techniques like classification, clustering, and different state-of-the-art machine learning models like ResNet.
- The platform used was Kaggle.

Winter School for Quantum Computing | IISER Kolkata

December 2022

- Introduction to basic quantum gates and quantum algorithms. Discussion of the issue of decoherence of quantum particles.
- Vast applications in the fields of medicine, finance, material sciences, etc.

Projects

Term Paper - Dark Matter Profiling

Feb 2025 - May 2025

- Discussed the effect of the Optical Spin Hall effect and spin-orbit interaction in the domain of metasurfaces.
- Mathematically derived the effects of the gradient of refractive indices in an anisotropic medium, which leads to the fabrication of metasurfaces.
- Finally, I wrote and presented a term paper on the field review of metasurfaces and flat optics.

Term Paper - Metasurfaces & Flat Optics

Oct 2024 - Dec 2024

- Discussed the effect of the Optical Spin Hall effect and spin-orbit interaction in the domain of metasurfaces.
- Mathematically derived the effects of the gradient of refractive indices in an anisotropic medium, which leads to the fabrication of metasurfaces.
- Finally, I wrote and presented a term paper on the field review of metasurfaces and flat optics.

Term Paper - Casimir Effect

January 2024 - May 2024

- We studied how quantum forces (Casimir forces) arise between two finite potential barriers in a 1D system using massless Dirac fermions.
- I used the Hellmann-Feynman theorem and matrix methods to calculate the force without running into the usual mathematical divergences.
- I showed how the force can switch from attractive to repulsive depending on the spin orientation of the barriers, which could be useful in nano-device design.

Library Management System

January 2024

- Developed a Library Management system with a user and an admin login in C.
- Incorporated various features for the book records, such as searching, issuing, deleting, etc.
- Received an outstanding group assignment mention for the project.

LiFi

2020

- Built a working prototype for transmitting audio data through the light spectrum.
- Won various awards for it at the school level.

Bus Booking Portal using Python and MySQL

[Github](#)

- Developed a bus booking portal using Python and MySQL as the DBMS.

Extra-Curricular

Summer Volunteer | [Ek Pehal foundation](#)

June 2023

- Volunteered to teach underprivileged middle and high school kids as part of the Ek Pehal Foundation.
- Taught English, Maths and Physics for 6 hours per week

Menstrual Hygiene Drive | [FARZ foundation](#)

November 2022

- Facilitated logistics and organised a menstrual hygiene campaign for schools in rural areas.
- Educated and lectured middle school students about the importance of menstrual hygiene in the modern day.
- Successfully helped organise the sanitary pad distribution drive for rural school students.

Additional Experience And Awards

- **Summer Fellow:** Completed a [Summer Research Fellowship Program 2024](#) offered by the [Indian Academy of Sciences](#) at SVNIT Surat.
- Part of the OPTICA community as a member of SPIE student chapter of IISER Kolkata.
- Represented my Institute as a part of the Football Team at Parakram - ISM Dhanbad.
- Participated and won 1st place in the Interbatch Football Tournament 2025.
- Participated and won the Spring Pong - annual Table Tennis auction tournament
- Won gold medal in the International English Olympiad.
- Participated as a guitarist in RAMPAGE - the annual inter-college Band tournament.
- Won gold medal in annual inter-college Futsal tournament.

Technological Skills

Coding: C, Python, MATLAB, Qiskit, MySQL, Latex.

Softwares: Quantum ESPRESSO, Origin, GNU-Plot, NanoHub(MD software), WIEN2k, Wolfram Mathematica.

Others: Unsupervised Machine Learning, Fundamentals of DFT and MD.